

EDUCATION

McGill University

Sept. 2023 - present

PhD Candidate in Mathematics and Statistics

- Supervisor: Dr. Elliot Paquette
- Research Topics: random matrix theory, stochastic processes, high-dimensional probability and optimization, generative models

McGill University

2020-2023

Master of Science in Mathematics and Statistics

- Supervisor: Dr. Adam Oberman
- Cumulative GPA: 4.0/4.0
- Associated with Mila – Quebec Artificial Intelligence Institute

The University of British Columbia

2015-2019

Bachelor of Science - Honours in Mathematics and Minor in Data Science

- Cumulative GPA: 4.33/4.33

RESEARCH EXPERIENCE

Microsoft + Nuance

2023 Summer

NLP Research Intern

- Implemented generative AI tools for customer service
- Augmented large language models using a knowledge base retriever
- Improve response quality with prompt engineering

Phyla

2021 Summer

Machine Learning Research Intern

- Developed a graph neural network for classification of irritable bowel diseases (IBD)
- Implemented a pipeline in PyTorch to run, test, and log experimental results
- Developed a custom cross validation method to reduce the impact of batch effects

The University of British Columbia

2020-2021

Research Assistant 2

- Analyzed and developed a differential equation model coupling the dynamics of opinion and disease
- Find that more reactive populations can experience smaller peak epidemic size, but more epidemic waves and more cases

Mathematics and Climate Research Network

2019-2021

Student Researcher

- Collaborated with an international group of students and professors to develop a novel data assimilation method
- Investigated the use of model and data dimension reduction methods to improve the effectiveness of particle filters

PUBLICATIONS

- Elliot Paquette, **Noah Marshall**, Lucas Benigni, Guangyuan Wang, Atish Agarwala, and Courtney Paquette. 2026. Phases of Muon: When Muon Eclipses SignSGD. *Preprint* (arXiv:2605.09552)
- Ke Liang Xiao, **Noah Marshall**, Atish Agarwala, and Elliot Paquette. 2025. Exact Risk Curves of signSGD in High-Dimensions: Quantifying Preconditioning and Noise-Compression Effects. *International Conference on Machine Learning*
- **Noah Marshall**, Ke Liang Xiao, Atish Agarwala, and Elliot Paquette. 2025. To Clip or Not to Clip: the Dynamics of SGD with Gradient Clipping in High-Dimensions. *International Conference on Learning Representations*
- Tiago Salvador, Stephanie Cairns, Vikram Voleti, **Noah Marshall**, and Adam Oberman. 2022. FairCal: Fairness Calibration for Face Verification. *International Conference on Learning Representations*
- Rebecca C. Tyson, **Noah D. Marshall**, and Bert O. Baumgaertner. 2022. Transient prophylaxis and multiple epidemic waves. *AIMS Mathematics*

- Aishah Albarakati, Marko Budišić, Rose Crocker, Juniper Glass-Klaiber, Sarah Iams, John Maclean, **Noah Marshall**, Colin Roberts, and Erik S. Van Vleck.* 2022. Model and Data Reduction for Data Assimilation: Particle Filters Employing Projected Forecasts and Data with Application to a Shallow Water Model. *Computers & Mathematics with Applications* (*equal contribution, sorted by last name)

TEACHING AND MENTORSHIP

Course Instructor

present

McGill University

- Instructor for Intermediate Calculus, including a current section

Directed Reading Program

2021-present

McGill University

- Lead a mentored reading program for undergraduate students
- Studied generative diffusion models and probabilistic graphical models
- Currently directing reading on tree indexed stochastic processes and percolations on neural networks

Graduate Teaching Assistant

2022-present

McGill University

- 2024: Introduction to stochastic processes
- 2023: Calculus 1
- 2022: Calculus 1

Undergraduate Teaching Assistant

2016-2019

The University of British Columbia

- Calculus 1
- Calculus 2
- Introduction to Probability

Supplemental Learning Instructor

2016-2019

The University of British Columbia

- Provide supplemental lectures for difficult first year math courses
- Prepare practice final exams and midterms
- Lead final exam and midterm review sessions

AWARDS

Canadian Graduate Scholarship (NSERC)

present

- Program provides financial support to high-calibre students

Institut des sciences mathématiques Graduate Scholarship

present

- Awarded to outstanding PhD students in mathematics

NSERC Undergraduate Student Research Award

2018

- Funding awarded to undergraduate students pursuing summer research in natural sciences or engineering